Ryvath Mattey

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Education

University of Georgia Athens, GA

B.S. Computer Science, Minor in Microbiology *Aug. 2023 – May 2027*

- **Relevant Coursework:** Data Structures, Algorithms, Software Development, Computer Architecture, Databases
- **Certifications:** AWS Certified Cloud Practitioner, HIPAA Certified

Technical Skills

Languages: Java, Python, Go, TypeScript, JavaScript, SQL (PostgreSQL), C/C++, HTML/CSS
Frameworks & Libraries: Spring Boot, React, FastAPI, gRPC, PyTorch, Scikit-Learn, HAPI FHIR, Claude API, Whisper (faster-whisper)
Infrastructure: Kafka, Redis, DynamoDB, Docker, Kubernetes, Terraform, AWS (EKS, RDS, Lambda, Cognito, KMS)
Tools & Testing: Git, GitHub Actions, Prometheus, Grafana, OpenTelemetry, Jaeger, Testcontainers, Playwright, pytest, k6, Linux

Projects

Chartwise: AI Clinical Documentation Pipeline | *Kafka, Go, Python* github.com/Rybro63/chartwise

- Built an event-driven pipeline that transcribes visits with local Whisper, drafts source-cited SOAP notes with the Claude API, and files approved notes to FHIR; deployed on AWS EKS (Terraform) with GitHub Actions CI (synthetic data only)
- Tuned transcription to 2.5% word error rate, 100% speaker attribution, and 45/47 drug names, at 12× real time
- Built a note-quality eval (answer-keyed visits plus an LLM judge): source citations halved unsupported sentences (0.9% → 0.4%) in Claude Opus drafts; Haiku hit 98.6% fact recall at $0.0015 vs. $0.041 per note
- Used a transactional outbox and idempotent Kafka consumers: a worker killed mid-stream lost or duplicated 0 of 200 notes
- Built a Go gRPC gateway with Redis request and LLM-token budgets, jittered retries, and a circuit breaker; in a simulated 60s LLM outage the breaker opened in 8s with 0 failed drafts
- Load-tested with k6 against a 30s SLO (100% of 1,029 drafts met it, fake LLM); a live Claude run exposed head-of-line blocking, and per-partition sliding commits cut p50 from 26.4s to 18.7s
- Wrote a safety check that caught 100% of 728 injected drug, allergy, and dose errors with 0 false positives on 544 clean drafts; envelope-encrypted PHI (AWS KMS) with an append-only audit log

batchserve: Dynamic Batching Inference Server | *Go, Concurrency* github.com/Rybro63/batchserve

- Built a request-batching inference server in Go (standard library only) that amortizes a fixed per-call backend cost across batched requests, raising goodput 12.8× (23 → 299 req/s) and eliminating timeouts at 300 req/s
- Implemented bounded-queue admission control, deadline dropping, and size-or-time flushing with distinct statuses for "never attempted" (503) vs. "backend failed" (500); validated with 44 tests under Go's race detector
- Showed queue depth dictates failure mode: a 16-slot queue shed load in microseconds and did 5.4× more useful work than a 1024-slot queue that timed out 3,271 requests

AI Symptom Checker (Full-Stack) | *React, FastAPI, AWS Lambda, DynamoDB, Cognito* live demo

- Built a multi-user AI triage app (React, FastAPI on AWS Lambda, Cognito JWT auth) using the Claude API across 4 triage levels, with a 0.35 confidence threshold that never withholds emergencies

Work Experience

Undergraduate Researcher – VIPR AI/ML and Robotics Aug. 2024 – May 2025

The University of Georgia *Athens, GA*

- Integrated robotic hardware with REST APIs and PyTorch in Python for real-time farm monitoring
- Benchmarked YOLOv5–v12 across five sizes: 98.4% mAP50 (YOLOv12x), 88% less training time (YOLOv8n)
- Automated RGB–depth alignment and dataset preprocessing in Python across 1,000+ poultry frames

Medical Scribe Intern Aug. 2023 – May 2024

Prime Health Medical Center *Buford, GA*

- Managed 30–40 patient records per shift with near-zero errors in QTC and Optum LHI EHR systems under HIPAA